

- The *power set of A* is denoted by $\mathcal{P}(A)$ and is given by,
 $\mathcal{P}(A) = \{X \mid X \text{ is a set and } X \subseteq A\}$
- The *universal set* is denoted by the letter U or E.

Set Operations

- The *union* $A \cup B$ of A and B is defined by

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

- The *intersection* $A \cap B$ of A and B is defined by

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

- The *set difference* $A \setminus B$ of A and B is the set

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

$A \setminus B$ is also called the *relative complement of B with respect to A*.

- If U is a given universal set, then $U \setminus A$ is just called the *complement of A*, written as A^c

$$A^c = \{x \mid x \in E \wedge x \notin A\}$$

Ex: Let A and B be two sets. Prove that:

- (i) $A \setminus A = \emptyset$.
- ✓(ii) $A \setminus B \subseteq A$. →
- (iii) $A \setminus \emptyset = A$.
- ✓(iv) $A \setminus B = \emptyset$ if and only if $A \subseteq B$.

Let $x \in E$ — (i)
 $x \in A \setminus B \Rightarrow x \in A \wedge x \notin B$
 $\Rightarrow x \in A$
 $\therefore x \in A \setminus B \Rightarrow x \in A$ — (ii)
 By (i), (ii) $\forall x \in E, x \in A \setminus B \Rightarrow x \in A$
 $\therefore A \setminus B \subseteq A$ □

Ex For subsets A and B , show that $(A \setminus B)^c = A^c \cup B$.

Proof of (ii): Let x be an arbitrary element — (i)

Suppose $x \in A \setminus B$ — (ii)

Then $x \in A$ and $x \notin B$

$\therefore x \in A$ — (iii)

By (i), (ii) and (iii) we see $\forall x, x \in A \setminus B \Rightarrow x \in A$

$\therefore A \setminus B \subseteq A$ □

Proof of (ii) by Method of Contradiction:

Let $A-B \subseteq A$ — (i)

Suppose (i) is false. — (ii)

i.e. $A-B \not\subseteq A$

$\therefore$ there exists $x_0 \in A-B$ such that $x_0 \notin A$.

$\therefore x_0 \in A-B$ and $x_0 \notin A$ for some x_0 .

$\therefore x_0 \in A$ and $x_0 \notin B$ and $x_0 \notin A$

$\therefore x_0 \in A$ and $x_0 \notin A$ is a contradiction.

$\therefore$ (i) is true

This complete the proof. $\square$

Proof of (iv) $\Rightarrow$:

Suppose $A-B = \emptyset$ — (i)

Let x be an arbitrary element — (ii)

Suppose $x \in A$. — (iii)

Suppose $x \notin B$

Then $x \in A-B$

i.e. $x \in \emptyset$ is a contradiction.

$\therefore x \in B$ — (ii)

$\therefore \forall x, x \in A \Rightarrow x \in B$
 $\therefore A \subseteq B$ //

Proof of (iv) $\Leftarrow$:

Suppose $A \subseteq B$ — (i)

clearly $\emptyset \subseteq A-B$ — (ii)

Let x be an arbitrary element — (1)

Assume that $x \in A-B$.

$\therefore x \in A$ and $x \notin B$.

$\therefore x \in B$ and $x \notin B$. (by (i))

$x \in \emptyset$

$\therefore A-B \subseteq \emptyset$

$\therefore A-B = \emptyset$ //

Note:

$$P \Leftrightarrow Q \equiv (P \Rightarrow Q \wedge Q \Rightarrow P)$$

$$\equiv (P \Rightarrow Q \wedge \sim P \Rightarrow \sim Q)$$

Proof of (iv) $\Leftarrow$: Suppose $A-B \neq \emptyset$

Then there exists at least x_0 such that $x_0 \in A-B$.

i.e. there exists x_0 such that $x_0 \in A$ and $x_0 \notin B$.
 $A \not\subseteq B$.

$$\therefore A - B \neq \emptyset \Rightarrow A \not\subseteq B$$

$$\text{Hence, } A \subseteq B \Rightarrow A - B = \emptyset$$

Proof of $(A - B)^c = A^c \cup B$:

Let x be an arbitrary element — (i)

Suppose $x \in (A - B)^c$ — (ii)

Then, $x \in E$ and $x \notin (A - B)$

i.e. $x \in E$ and $\neg(x \in A - B)$

i.e. $x \in E$ and $\neg(x \in A \wedge x \notin B)$

$x \in E$ and $(x \notin A \vee x \in B)$

$\therefore (x \in E \wedge x \notin A) \vee (x \in E \wedge x \in B)$

$\therefore x \in A^c \vee x \in B$

$\therefore x \in A^c \cup B$ — (iii)

By (i), (ii), (iii) $\forall x, x \in (A - B)^c \Rightarrow x \in A^c \cup B$ ✓

Ex:

Proof of $(A - B)^c = A^c \cup B$: Let $x \in E$ — (i)

$$x \in (A - B)^c \Leftrightarrow x \in E \wedge x \notin (A - B)$$

$$\Leftrightarrow x \in E \wedge \neg(x \in A - B)$$

$$\Leftrightarrow x \in E \wedge \neg(x \in A \wedge x \notin B)$$

$$\Leftrightarrow x \in E \wedge (x \notin A \vee x \in B)$$

$$\Leftrightarrow (x \in E \wedge x \notin A) \vee (x \in E \wedge x \in B)$$

$$\Leftrightarrow x \in A^c \vee x \in B$$

$$\Leftrightarrow x \in A^c \cup B$$

$$\therefore x \in (A - B)^c \Leftrightarrow x \in A^c \cup B \text{ — (ii)}$$

$$\text{By (i), (ii)} \quad \forall x \in E, x \in (A - B)^c \Leftrightarrow x \in A^c \cup B$$

$$\therefore (A - B)^c = A^c \cup B$$

3.3 Set Identities

Identity Laws

$$\begin{aligned} * A \cup \emptyset &= A \\ A \cap U &= A \end{aligned}$$

Domination Laws

$$\begin{aligned} A \cup U &= U \\ A \cap \emptyset &= \emptyset \end{aligned}$$

De Morgan's Laws

$$\begin{aligned} * (A \cup B)^c &= A^c \cap B^c * \\ (A \cap B)^c &= A^c \cup B^c \end{aligned}$$

Commutative Laws

$$\begin{aligned} A \cup B &= B \cup A * \\ A \cap B &= B \cap A \end{aligned}$$

Associative Laws

$$\begin{aligned} A \cup (B \cap C) &= (A \cup B) \cap C = A \cup B \cap C * \\ A \cap (B \cup C) &= (A \cap B) \cup C = A \cap B \cup C \end{aligned}$$

Absorption Laws

$$\begin{aligned} * A \cup (A \cap B) &= A \\ A \cap (A \cup B) &= A \end{aligned}$$

Idempotent Laws

$$\begin{aligned} A \cup A &= A \\ A \cap A &= A * \end{aligned}$$

Distributive Laws

$$\begin{aligned} A \cap (B \cup C) &= (A \cap B) \cup (A \cap C) * \\ A \cup (B \cap C) &= (A \cup B) \cap (A \cup C) \end{aligned}$$

Complement Laws

$$\begin{aligned} A \cup A^c &= U \\ A \cap A^c &= \emptyset \end{aligned}$$

Complementation Law

$$(A^c)^c = A$$

Proofs using Set Identities

Suppose, E is a universal set. Now consider the following seven results for any subsets A, B, C of E .

- R1: $A \cup \emptyset = A$
- R2: $A \cap A^c = \emptyset$
- R3: $(A^c)^c = A$
- R4: $A \cap B = B \cap A$
- R5: $(A \cap B)^c = A^c \cup B^c$
- R6: $A \setminus B = A \cap B^c$
- R7: $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$

Now, using only seven results and the properties of "=", prove algebraically the following for arbitrary subsets A, B, C of E .

- (α) $(B \setminus A) \cup (C \setminus A) = (B \cup C) \setminus A$
- (β) $B^c \setminus (B^c \setminus A) = B^c \cap A$

$$\text{Let } x \in E \text{ --- (i)}$$

$$\begin{aligned} x \in A \cup \emptyset &\iff x \in A \vee x \in \emptyset \\ &\iff x \in A \vee F \\ &\iff x \in A \end{aligned}$$

$$\therefore x \in A \cup \emptyset \iff x \in A \text{ --- (ii)}$$

$$\therefore A \cup \emptyset = A \quad \square$$

$$\text{Let } x \in E \text{ --- (i)}$$

$$\begin{aligned} x \in A \cup B &\iff x \in A \vee x \in B \\ &\iff x \in B \vee x \in A \\ &\iff x \in B \cup A \end{aligned}$$

$$\therefore x \in A \cup B \iff x \in B \cup A \text{ --- (ii)}$$

$$\text{by (i), (ii)} \quad A \cup B = B \cup A \quad \square$$

$$(Ex)$$

$$\text{Proof: Let } x \in E \text{ --- (i)}$$

$$\begin{aligned} x \in A \cap (B \cup C) &\iff x \in A \wedge x \in B \cup C \\ &\iff x \in A \wedge (x \in B \vee x \in C) \end{aligned}$$

$$\iff (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C)$$

$$\iff (x \in A \cap B) \vee (x \in A \cap C)$$

$$\iff x \in (A \cap B) \cup (A \cap C)$$

$$\therefore x \in A \cap (B \cup C) \iff x \in (A \cap B) \cup (A \cap C) \text{ --- (ii)}$$

$$\text{by (i), (ii)} \quad \forall x \in E, x \in A \cap (B \cup C) \iff x \in (A \cap B) \cup (A \cap C)$$

$$\therefore A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad \square$$

$$\begin{aligned} P &\equiv P \vee F \\ P &\equiv P \wedge T \end{aligned}$$

$$P \vee Q \equiv Q \vee P$$

End of Dis